

SAM PETER

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Introduction

Driven by curiosity and a constant desire to learn, I challenge myself beyond limits to gain knowledge, build meaningful solutions, and grow through every experience. Passionate about AI/ML, Python, and web application development, I enjoy transforming ideas into practical and impactful software solutions. Adaptable and problem-oriented, I continuously explore emerging technologies while strengthening my technical and analytical skills.

Education

NSS College of Engineering, Palakkad

Expected May 2027

B.Tech, Electronics and Communication Engineering — CGPA: 6.84/10.00

Kendriya Vidyalaya (CBSE)

March 2023

Higher Secondary Education (Class XII)

Experience

AI & ML Intern, Novola.ai & AccelerateX

June 2025

- Built AI-based projects including Routine Setting Agent and Hand Gesture Recognition System.
- Worked on ML models such as Linear Regression, Logistic Regression, Decision Trees, and Random Forest.
- Explored computer vision, feature optimization, model training, and dataset preprocessing.

Positions of Responsibility

- **Secretary, IEEE PES SBC NSSCE (2026 – Present):** Organized Nexotech IEEE Conference 2026 and Luminous Logic workshop.
- **Operations Lead, iEDC NSSCE (2026 – Present):** Coordinated District Level YIP 7.0, YIP 8.0 and Kerala Innovation Festival 2025.

Projects

AI Fish Tank Simulator

- Engineered an interactive virtual space with real-time gesture tracking and low-latency rendering.
- **Tools Used:** Python, OpenCV, MediaPipe, Pygame, NumPy, gTTS

8-Bit Microprocessor Design

- Simulated an 8-bit microprocessor architecture featuring a modular Control Unit, Register File, and a 24+ op ALU.
- **Tools Used:** Verilog HDL, Xilinx Vivado, Digital Logic Design, Behavioral Simulation

Smart Automated Delivery Box

- Created an automated box with secure OTP verification, load-cell calibration, and real-time UART-to cloud updates.
- **Tools Used:** Python, Embedded C, UART, Telepot API, IoT Protocols, Hardware-Software Integration

Water Taranga – IoT Monitoring System

- Developed an IoT-based system for monitoring water and electricity usage with threshold-based alerts.
- **Tools Used:** Python, Flask/Django, IoT, HTML, CSS, JavaScript, MySQL

Certifications

- The Joy of Computing using Python - IIT Madras, NPTEL 2025
- Deep Learning - IIT Ropar, NPTEL 2026

Achievements

- Participated in the national-level Bharatiya Antariksh Hackathon 2025.
- Presented the Smart Automated Delivery Box System at the YIP District Evaluation panel.
- Developing Persistent Scene Change Detection Pipeline for Vehant Hackathon 2026.
- 2nd Runner-Up of CPL with Team Tridents